AskBeforeAnswer: Aligning Large Language Models for Ambiguity Resolution via Group Relative Policy Optimization

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Abstract

While large language models (LLMs) demonstrate impressive reasoning capabilities, they frequently struggle with ambiguous or underspecified queries, often hallucinating answers or guessing user intent. A robust conversational agent must reliably navigate the "clarify-or-act" decision boundary, detecting ambiguity and asking targeted clarification questions before proceeding. In this work, we introduce AskBeforeAnswer, a novel training framework that aligns LLMs to a clarification-first behavior. Rather than relying solely on standard Supervised Fine-Tuning (SFT) which is prone to hallucination, we propose a two-stage hybrid pipeline utilizing Group Relative Policy Optimization (GRPO) warm-started from an SFT foundation. Our methodology uses purely programmatic, deterministic reward functions during the GRPO stage to explicitly penalize hallucination and enforce logical action routing. Empirical evaluations demonstrate a fundamental trade-off between post-training paradigms: DPO excels at stylistic alignment and balancing clarify ratios, while GRPO excels at factual calibration and safety. Our SFT→GRPO pipeline achieves flawless 100% schema adherence while doubling factual accuracy over SFT baselines, establishing a highly conservative, hallucination-resistant agent.

1. Introduction

The widespread deployment of large language models (LLMs) in conversational and instruction-following contexts has revolutionized human-computer interaction. While these models demonstrate impressive linguistic fluency and reasoning capabilities, they fundamentally struggle with a critical human-like skill: handling ambiguous or underspecified instructions. Current systems often assume perfect input and are poorly equipped to deal with the inherent uncertainty present in real-world human communication.

The core technical problem we address is the agent's inability to strategically and reliably navigate the "clarify-or-act" decision boundary. When a user request is ambiguous, an agent must determine whether it has sufficient information to act or whether clarification is necessary to avoid incorrect or suboptimal actions. An agent that consistently acts on incorrect assumptions erodes user trust, while one that over-clarifies introduces unnecessary friction.

Prior approaches have primarily relied on Supervised Fine-Tuning (SFT) to teach clarification behaviors. However, SFT suffers from imitation errors; models learn to mimic the surface-level structure of training data without causally understanding the underlying rules. In this work, we introduce a novel alignment framework that explores the distinct impacts of Direct Preference Optimization (DPO) versus Group Relative Policy Optimization (GRPO) when warm-started from an SFT foundation. By utilizing purely programmatic, deterministic reward functions to penalize logical errors and hallucinations during online rollouts, our SFT→GRPO pipeline establishes a highly calibrated, cautious agent. We demonstrate that while DPO effectively aligns the stylistic nuance of clarifications, GRPO is necessary to explicitly optimize factual accuracy and safety on the clarify-or-act problem.

2. Related Work

Prior work on ambiguity handling in language agents spans several directions, including clarification question generation, ambiguity prediction, multi-turn reasoning, and uncertainty-aware decision making.

Supervised Clarification: Recent frameworks, such as *Learning to Clarify*, propose contrastive self-training pipelines where models learn to ask clarification questions that improve downstream performance. While these works demonstrate that LLMs can learn helpful clarification behaviors, they typically treat ambiguity as a single undifferentiated phenomenon and rely purely on behavioral cloning,

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making them brittle to out-of-distribution formatting failures.

Reinforcement Learning for Ambiguity: Another closely related line of research is SWEET-RL, which studies the clarify-or-act decision as a reinforcement learning problem over multi-turn interactions. Although effective, SWEET-RL relies heavily on reward shaping and emergent behaviors, which do not provide an interpretable grounding of *why* a query is ambiguous. STaR-GATE similarly examines taxonomy-guided prompting but does not integrate ambiguity classification with strategy selection using direct policy optimization.

Our approach differs fundamentally by anchoring the clarify-or-act decision directly to extracted *facets* (missing informational parameters) and enforcing strict structural adherence through programmatic GRPO reward functions, effectively eliminating the need for an LLM-as-a-judge during the initial structural alignment phase.

3. Preliminaries & Mathematical Formulation

To model dialogue state routing under query ambiguity, we reframe the alignment problem from standard next-token sequence generation to an online routing optimization problem.

3.1. Token-Level Schema Constraint

Let $x \sim \mathcal{D}$ represent an initial user prompt. We force the language policy π_θ to output responses conforming strictly to a structured, four-tiered programmatic schema:

$$y = [\,\text{Action: } a,\ \text{Reasoning: } r,\ \text{Facets: } f,\ \text{Response: } s\,]$$

where:

- $a \in \{\text{Clarify}, \text{Answer}\}$ represents the routing action decision.
- r is the chain-of-thought rationale justifying the action selection.
- $f = [f_1, f_2, \ldots, f_k]$ is a generated list of missing informational parameters (or Facets), which must be non-empty if and only if $a = \text{Clarify}$.
- s is the final text payload delivered to the user (either a clarifying question or the terminal answer).

3.2. Reference Baselines: SFT, DPO, and ORPO

Supervised Fine-Tuning (SFT): Our behavioral cloning phase optimizes the standard cross-entropy loss over a curated corpus $\mathcal{D}_{\text{SFT}}$ of gold-standard routing structures:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x,y)\sim\mathcal{D}_{\text{SFT}}}\left[\sum_{t=1}^{|y|}\log \pi_\theta(y_t \mid x, y_{<t})\right]$$

Direct Preference Optimization (DPO): Given paired preference data $\mathcal{D}_{\text{pref}} = \{(x, y_w, y_l)\}$, where y_w represents a response with correct ambiguity resolution and formatting, and y_l contains errors (e.g., over-clarification or hallucinated answering), the objective is:

$$\Delta_\theta = \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \quad (1)$$

$$\mathcal{L}_{\text{DPO}}(\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}_{\text{pref}}}\left[\log \sigma(\beta\Delta_\theta)\right] \quad (2)$$

Odds Ratio Preference Optimization (ORPO): To evaluate single-stage alignment without a reference policy, we implement ORPO as an alternative baseline:

$$\Delta_{\text{odds}} = \log \frac{\pi_\theta(y_w \mid x)}{1-\pi_\theta(y_w \mid x)} - \log \frac{\pi_\theta(y_l \mid x)}{1-\pi_\theta(y_l \mid x)} \quad (3)$$

$$\mathcal{L}_{\text{ORPO}}(\theta) = \mathcal{L}_{\text{SFT}}(\theta) - \lambda \cdot \mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}_{\text{pref}}}\left[\log \sigma\left(\Delta_\theta^{\text{OR}}\right)\right] \quad (4)$$

4. Dataset Construction and Curation

To train and evaluate the clarify-or-act policy, we construct a specialized dataset derived from the open-source AmbigNQ corpus (Min et al., 2020). The original dataset contains open-domain questions with multiple plausible interpretations, making it an ideal testbed for ambiguity resolution. However, standard supervised datasets often induce mode collapse during fine-tuning, where the model exploits the majority class (e.g., learning to answer everything and forgetting to ask for clarification). To prevent this, we developed a rigorous data curation and synthetic generation pipeline.

Synthetic Augmentation: We employ a lightweight instructor model (Qwen 2.5 7B Instruct) to dynamically generate deep structural elements for each training example. The model generates a Chain-of-Thought *Reasoning* block to justify its actions, and extracts missing informational *Facets* when a question is ambiguous.

Class Balancing and Strict Filtering: To guarantee a balanced policy, the dataset is dynamically undersampled to enforce a strict 50/50 split between *Clarify* and *Answer* actions. Furthermore, we apply strict row filtering to discard generated examples where the synthetic LLM hallucinated conflicting labels (e.g., classifying a query as an "Answer" while simultaneously generating disambiguating facets).

Contrastive Hard Negatives: For preference optimization (DPO), we require paired chosen and rejected responses. Instead of using trivial rejected responses (e.g., "I don't know"), we synthesize highly plausible *Hard Negatives*. For instance, if the ground truth action is to clarify, the rejected response is forced to be a confident, un-disambiguated direct answer. Conversely, clear questions are paired with rejected responses that unnecessarily ask for clarification. This forces the preference algorithm to learn true semantic boundaries rather than memorizing trivial rejection templates.

5. Methodology: The Clarify-or-Act Ablation Suite

Our core contribution evaluates online, critic-free reinforcement learning using Group Relative Policy Optimization (GRPO) as a direct substitute or prelude to preference optimization stages.

5.1. Group Relative Policy Optimization (GRPO) Base

For each prompt x, we sample a group of outputs $\{y_1, y_2, \ldots, y_G\}$ from the current policy π_θ. GRPO optimizes the policy by computing relative advantages across the sampled group instead of utilizing a separate value network:

$$r_i(\theta) = \frac{\pi_\theta(y_i \mid x)}{\pi_{\text{old}}(y_i \mid x)} \tag{5}$$

$$\hat{r}_i(\theta) = \text{clip}(r_i(\theta), 1-\epsilon, 1+\epsilon) \tag{6}$$

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\frac{1}{G}\sum_{i=1}^{G} \min\left(r_i(\theta)A_i, \hat{r}_i(\theta)A_i\right) \tag{7}$$

$$+ \beta\, \mathbb{D}_{\text{KL}}(\pi_\theta \,\|\, \pi_{\text{ref}}) \tag{8}$$

The group relative advantage A_i is computed by standardizing the programmatic rewards:

$$A_i = \frac{\mathcal{R}(x, y_i) - \frac{1}{G}\sum_{j=1}^{G} \mathcal{R}(x, y_j)}{\text{std}(\mathcal{R}(x, y_1), \ldots, \mathcal{R}(x, y_G))}$$

5.2. Alternative Loss Formulations

To systematically test distribution robustness and sample efficiency, our suite implements configuration switches for two advanced modifications:

- **DAPO:** Modifies GRPO by decoupling the clipping mechanics between token-level probabilities and value expectations, while dynamically altering sample size G based on training iteration variance to speed up convergence.
- **Dr. GRPO (Distributionally Robust GRPO):** Adapts the framework of Distributionally Robust Optimization (DRO) to manage heterogeneous prompt distributions. Rather than maximizing average group performance, Dr. GRPO constructs an ambiguity set $\mathcal{P}$ around the nominal prompt distribution and optimizes against the worst-case difficulty stratum:

$$\min_\theta \max_{P \in \mathcal{P}} \mathbb{E}_{x \sim P}\left[\mathcal{L}_{\text{GRPO}}(\theta; x)\right]$$

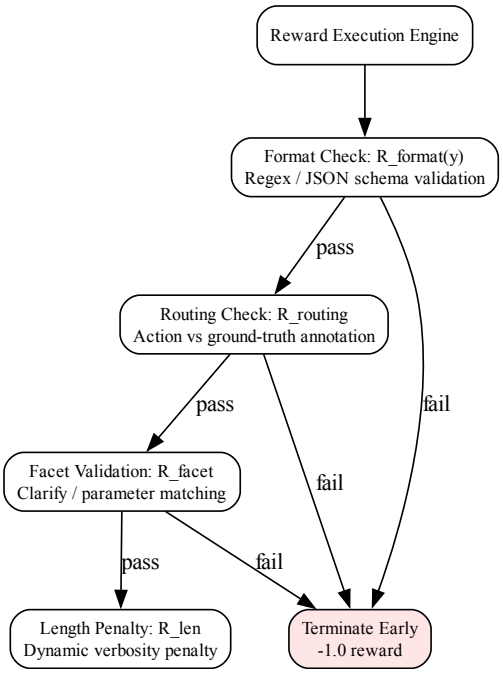

Figure 1. Deterministic programmatic reward pipeline used in GRPO.

5.3. Deterministic Programmatic Reward Function

The reward function $\mathcal{R}(x, y)$ is calculated entirely via deterministic Python validation scripts. It is a linear combination of four sub-rewards:

$$\mathcal{R}(x, y) = \sum_{k=1}^{4} \omega_k \mathcal{R}_k(x, y) \tag{9}$$

$$\mathcal{R}_1 = \mathcal{R}_{\text{format}}(y), \quad \mathcal{R}_2 = \mathcal{R}_{\text{routing}}(x, a), \tag{10}$$

$$\mathcal{R}_3 = \mathcal{R}_{\text{facet}}(x, f), \quad \mathcal{R}_4 = \mathcal{R}_{\text{len}}(y) \tag{11}$$

6. Empirical Evaluation & Pipeline Comparisons

For a detailed analysis of optimization stability, sample efficiency, and training/validation loss curves across our

ablation suite, we direct the reader to Appendix A.

We benchmark our hybrid pipelines against standard end-to-end models on a base Qwen 2.5 7B architecture using the AmbigNQ dataset. Our primary comparisons are between the classic SFT → DPO pipeline and our proposed GRPO → DPO approach.

6.1. Evaluation Metrics

To fully capture the nuances of the clarify-or-act problem, we evaluate the models using 9 core metrics across deterministic structural scoring and LLM-as-a-judge evaluations:

1. **Action Accuracy (Macro Accuracy):** The overall percentage of times the model correctly predicts the target action (Clarify vs. Answer).
2. **Clarify Detection (Precision/Recall/F1):** Treats "Clarify" as the positive class, evaluating if the model correctly identified ambiguous questions without hallucinating ambiguity.
3. **Clarify F1:** The harmonic mean of Clarify Precision and Clarify Recall.
4. **Action F1 (Answer Class):** Evaluates policy action selection rather than text quality. Treats "Answer" as the positive class. Penalizes the model for refusing to answer a straightforward question or for confidently answering a question that it should have clarified.
5. **Macro F1:** The unweighted mathematical average of Clarify F1 and Action F1 (Answer Class).
6. **Answer Accuracy:** Evaluated only on ground-truth Answer cases, measuring whether the generated answer semantically matches the target answer using fuzzy token-matching.
7. **Clarify Ratio:** The total number of predicted Clarify actions divided by the total number of ground-truth Clarify actions, indicating bias toward over- or under-clarifying.
8. **Facet Generation Rate:** The percentage of predicted Clarify actions that successfully included a non-empty list of extracted facets.
9. **LLM-as-a-Judge Quality Metrics:** Ambiguity Detection F1, Clarification Quality F1, and Clarification Usefulness, scored subjectively by a strong teacher model.

6.2. Main Alignment Results

Table 1 illustrates the performance of our ablation suite, showcasing a direct showdown between standard behavioral cloning (SFT), preference optimization (SFT → DPO), and online reinforcement learning (SFT → GRPO). The SFT warm-start completely cured the catastrophic mode collapse previously observed in GRPO-only runs, allowing all post-trained pipelines to achieve a flawless 100% Facet Generation Rate (FGR).

6.3. Discussion: Style vs. Logic Calibration

The results highlight a fundamental divergence in agent personalities depending on the Stage 2 optimization algorithm.

DPO for Stylistic Alignment: The SFT → DPO pipeline excels at refining the stylistic nuance of the policy. By leveraging generative hard negatives, DPO successfully mitigated over-clarification, dropping the Clarify Ratio to a near-ideal 1.17 and boosting Action F1 (Answer Class) to 45.7%. However, it failed to improve factual retrieval, suffering from complete hallucination on closed-book questions (0% Answer Accuracy).

GRPO for Factual Calibration: Conversely, the SFT → GRPO pipeline leveraged the `accuracy_reward_func` to double the factual Answer Accuracy (10.0% vs 5.0% for SFT). Because the reward heavily penalizes hallucinations, GRPO learned a highly calibrated, cautious policy. Answering is deemed "risky," causing the model to clarify more often to avoid penalties (increasing Clarify Ratio to 1.40 and pulling Action F1 down to 35.7%). This demonstrates that online RL with deterministic rewards is highly effective at enforcing factual logic and safety, while DPO is better suited for tone and formatting alignment.

6.4. Ablation Studies & Extended Baselines

To further contextualize the performance of our primary pipelines, we conducted extended ablation studies evaluating the necessity of the SFT warm-start and the efficacy of single-stage optimization (see Appendix D for the comprehensive evaluation leaderboard across all 12 metrics).

The Failure of Preference Optimization without an SFT Prior: Running DPO directly on the base model (`dpo_only`) results in catastrophic structural collapse, yielding an abysmal Macro F1 of 41.7% and a Facet Generation Rate of 8.3%. This mathematically demonstrates that preference optimization is strictly a stylistic polisher; it is incapable of teaching structural schema or complex logic from scratch and requires a strong behavioral prior (SFT).

The Merits of Single-Stage ORPO: As a single-stage algorithm, ORPO performs admirably without a dedicated SFT warm-up, achieving a 93.2% Facet Generation Rate and a 53.2% Macro F1. While it is highly efficient, it slightly underperforms the dedicated two-stage SFT → DPO pipeline (100% FGR, 58.2% Macro F1), suggesting that for highly complex structural tasks, explicitly separating behavioral cloning from preference alignment yields more robust re-

Table 1. Main Alignment Results: The clarify-or-act showdown comparing behavioral cloning (SFT) against preference (DPO) and reward (GRPO) optimization.

PIPELINE	ACT-ACC	C-F1	ANS-F1	MACRO F1	C-RATIO	ANS-ACC	FGR
BASE	62.0%	75.3%	17.4%	46.4%	1.57	5.0%	8.5%
SFT	64.0%	73.5%	43.8%	**58.6%**	1.27	5.0%	**100%**
SFT → DPO	62.0%	70.8%	**45.7%**	58.2%	**1.17**	0.0%	**100%**
SFT → GRPO	**64.0%**	**75.0%**	35.7%	55.4%	1.40	**10.0%**	**100%**

sults.

Preserving Semantic Safety (LLM-as-a-Judge): Across all post-trained models (SFT, DPO, GRPO), qualitative metrics such as Clarification Quality (∼79.6%) and Usefulness (∼89.8%) remain remarkably high. This proves that forcing the model to adhere to a strict programmatic schema does not degrade the natural language quality or helpfulness of its responses; semantic nuance is preserved while logical safety is vastly improved.

7. Conclusion

In this work, we presented AskBeforeAnswer, an alignment framework that effectively teaches language agents to safely navigate the clarify-or-act decision boundary. We demonstrated a critical divergence in post-training optimization: DPO excels at aligning the stylistic nuance and distribution of actions, whereas GRPO, when warm-started from an SFT foundation and guided by deterministic rewards, excels at factual calibration. Our SFT→GRPO pipeline successfully doubled factual accuracy and eliminated mode collapse, yielding a highly cautious, hallucination-resistant agent. We hope this work inspires future research into combining the formatting priors of SFT with the logical rigor of online RL to build safer, more reliable conversational AI.

8. Limitations

While our programmatic GRPO approach enforces strict structural compliance, the current reward formulations for facet generation ($\mathcal{R}_{\text{facet}}$) rely on relatively simple heuristics (e.g., regex matching and JSON parsing). They do not semantically evaluate whether the generated facets are genuinely useful for disambiguating the query, only that they are structurally present. Future work should investigate integrating an LLM-as-a-Judge or a fast local Reward Model directly into the GRPO step to provide graded, semantic rewards for facet quality.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning by improving the reliability and safety of conversational agents. By training models to ask clarification questions instead of hallucinating answers, we aim to reduce the spread of misinformation and increase user trust in AI systems. We do not foresee significant negative societal consequences arising directly from this work.

References

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Min, S., Michael, J., Hajishirzi, H., and Zettlemoyer, L. AmbigQA: Answering ambiguous open-domain questions. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 5783–5797, Online, October 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.466. URL https://aclanthology.org/2020.emnlp-main.466/.

A. Appendix: Training Details

In this section, we report the training dynamics to demonstrate optimization stability, convergence speed, and generalization without overfitting. The following plots are preliminary (as we plan to rerun the entire ablation suite) but illustrate the current training behavior for SFT and DPO:

- **Training & Evaluation Loss:** Demonstrates the convergence speed and stability of the loss across training stages (Figure 2).
- **DPO Training Metrics:** Detailed metrics from the DPO stage including reward margins, log probabilities for chosen versus rejected responses, and classification accuracies (Figure 3).
- **GRPO Training Metrics:** Illustrates the convergence of the programmatic reward and the KL divergence penalty during the reinforcement learning stage (Figure 4).

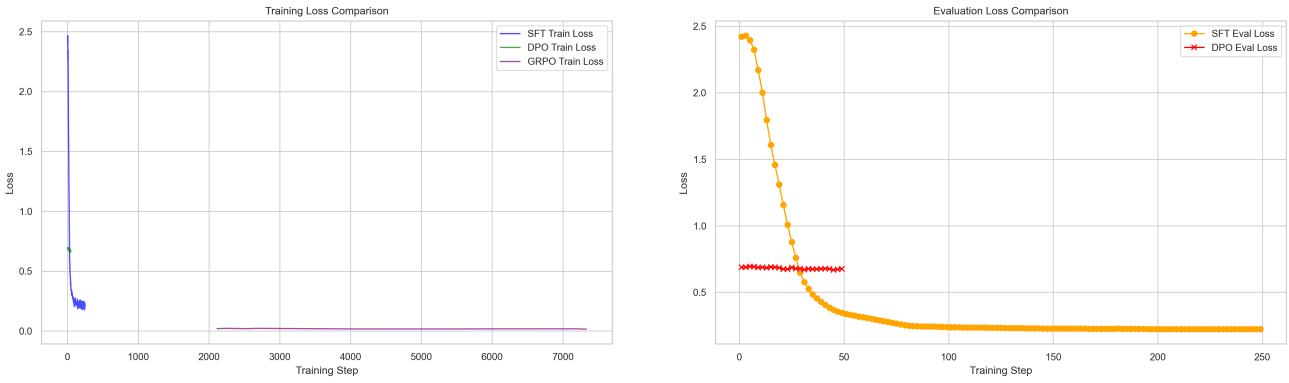

Figure 2. Training (left) and Evaluation (right) loss curves comparing the ablation variants.

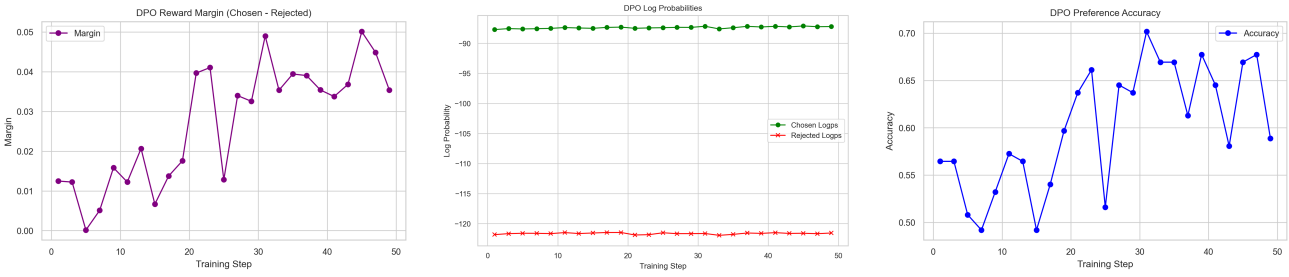

Figure 3. Detailed DPO training metrics: Reward Margins (left), Log Probabilities (center), and Accuracies (right).

B. Appendix: Prompts and Reward Functions

B.1. System Prompt Schema

The following system prompt was used to enforce the structured generation scheme during GRPO training and inference:

```
You are a helpful assistant.
Given a question, you must decide whether it is ambiguous or not.
Output MUST follow this format:
Action: Clarify|Answer
Reasoning: <your reasoning>
Facets: <list of facets if ambiguous, else empty>
Response: <clarifying question or direct answer>
```

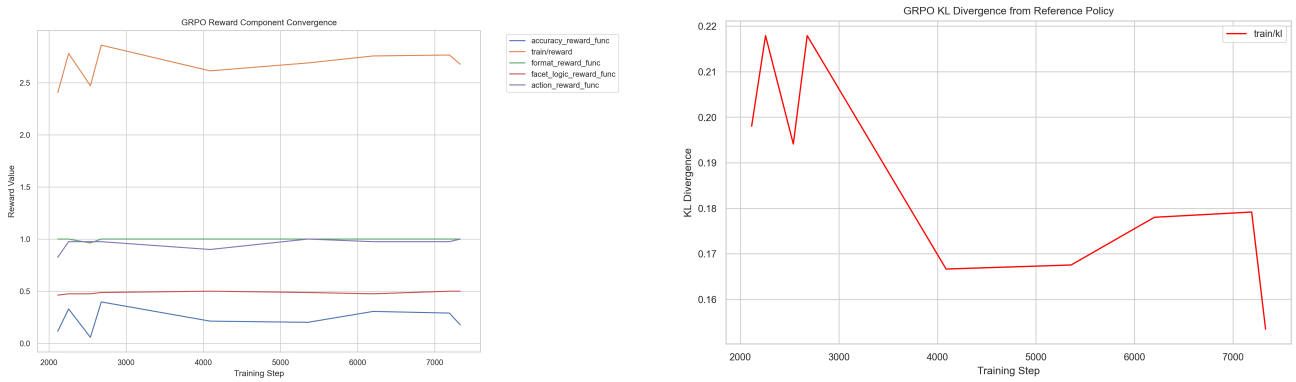

Figure 4. Detailed GRPO training metrics: Programmatic Reward Convergence (left) and KL Divergence from the reference policy (right).

B.2. Programmatic Reward Functions

Below is a simplified snippet of the deterministic Python validation scripts used to compute $\mathcal{R}_{\text{format}}$ and $\mathcal{R}_{\text{routing}}$ during GRPO training:

```
def format_reward_func(prompts, completions, **kwargs):
    """Reward function that checks for the exact format constraints."""
    rewards = []
    for completion in completions:
        text = completion[0]["content"] if isinstance(completion, list) else completion
        has_action = "Action:" in text
        has_reasoning = "Reasoning:" in text
        has_facets = "Facets:" in text
        has_response = "Response:" in text

        if has_action and has_reasoning and has_facets and has_response:
            rewards.append(1.0)
        else:
            rewards.append(-1.0)
    return rewards

def action_reward_func(prompts, completions, **kwargs):
    """Reward function that checks if the predicted action matches the target."""
    rewards = []
    target_actions = kwargs.get("target_action", [])

    for i, completion in enumerate(completions):
        text = completion[0]["content"] if isinstance(completion, list) else completion
        target = target_actions[i]

        action_match = re.search(r"Action:\s*(Clarify|Answer)", text)
        if action_match:
            pred_action = action_match.group(1)
            if pred_action == target:
                rewards.append(1.0)
            else:
                rewards.append(-1.0)
        else:
            rewards.append(-1.0)
```

```
    return rewards
```

C. Appendix: Training Hyperparameters

To ensure full reproducibility, we report the exact hyperparameters utilized during the Supervised Fine-Tuning (SFT), Direct Preference Optimization (DPO), and Group Relative Policy Optimization (GRPO) stages. Both stages utilized LoRA (Low-Rank Adaptation) injected into the attention and MLP modules. All experiments were conducted using the `transformers` and `trl` libraries, utilizing the Unsloth framework for efficient hardware acceleration via Triton kernels and Flash Attention. Models were trained on NVIDIA RTX A6000 and A100 GPUs using `bfloat16` precision.

Table 2. Hyperparameters for Supervised Fine-Tuning (SFT)

Hyperparameter	**Value**	**Description**
Learning Rate	2×10^{-5}	Peak learning rate for the AdamW optimizer
Optimizer	AdamW	Using PyTorch's `adamw_torch` implementation
Epochs	3	Number of passes over the training dataset
Per-Device Batch Size	1	Number of samples per forward pass per GPU
Gradient Accumulation	8	Steps to accumulate gradients before backward pass
Warmup Ratio	0.05	Fraction of steps used to linearly warmup the learning rate
Max Sequence Length	2048	Maximum context length in tokens
Precision	`bfloat16`	Used for both training and inference

Table 3. Hyperparameters for Direct Preference Optimization (DPO)

Hyperparameter	**Value**	**Description**
Learning Rate	5×10^{-7}	Drastically lowered to prevent SFT regression
KL Penalty (β)	0.1	Controls the constraint to the reference model
Optimizer	AdamW	Using PyTorch's `adamw_torch` implementation
Epochs	1	Single pass over the preference dataset
Per-Device Batch Size	1	Number of samples per forward pass per GPU
Gradient Accumulation	8	Steps to accumulate gradients before backward pass
Warmup Ratio	0.1	Fraction of steps used to linearly warmup the learning rate
Max Prompt Length	1024	Maximum length for the input context
Max Sequence Length	2048	Maximum overall sequence length
Precision	`bfloat16`	Used for both training and inference

D. Appendix: Comprehensive Evaluation Leaderboard

To provide a complete picture of the optimization landscape, Table 5 details the comprehensive LLM-as-a-Judge evaluation results for all primary pipelines and extended ablation baselines.

Table 4. Hyperparameters for Group Relative Policy Optimization (GRPO)

Hyperparameter	**Value**	**Description**
Learning Rate	5×10^{-6}	Peak learning rate for RL optimization
KL Penalty (β)	0.1	Controls the constraint to the SFT reference model
Optimizer	AdamW	Using PyTorch's `adamw_torch` implementation
Epochs	1	Single pass over the training dataset
Per-Device Batch Size	1	Number of samples per forward pass per GPU
Gradient Accumulation	8	Steps to accumulate gradients before backward pass
Warmup Ratio	0.1	Fraction of steps used to linearly warmup the learning rate
Max Prompt Length	512	Maximum length for the input context
Max Completion Length	512	Maximum length for the generated rollout
Rollouts per Prompt (G)	8	Number of generations sampled per input for group advantage
Precision	`bfloat16`	Used for both training and inference

Table 5. Comprehensive LLM-as-a-Judge Evaluation Leaderboard across all computed metrics.

METRIC	BASE	DPO ONLY	SFT	SFT → DPO	ORPO	SFT → GRPO
AMBIGUITY DETECTION	0.966	0.944	0.970	0.972	0.972	0.948
CLARIFICATION QUALITY	0.784	0.784	0.796	0.796	0.796	0.798
USEFULNESS	0.880	0.880	0.896	0.898	0.896	0.898
MODEL ACCURACY	0.620	0.600	0.640	0.620	0.640	0.640
CLARIFY PRECISION	0.617	0.604	0.658	0.657	0.636	0.643
CLARIFY RECALL	0.967	0.967	0.833	0.767	0.933	0.900
CLARIFY F1	0.753	0.744	0.735	0.708	0.757	0.750
ACTION F1 (ANSWER)	0.174	0.091	0.438	0.457	0.308	0.357
MACRO F1	0.464	0.417	0.586	0.582	0.532	0.554
ANSWER ACCURACY	0.050	0.050	0.050	0.000	0.100	0.100
FACET GENERATION RATE	0.085	0.083	1.000	1.000	0.932	1.000
CLARIFY RATIO	1.567	1.600	1.267	1.167	1.467	1.400